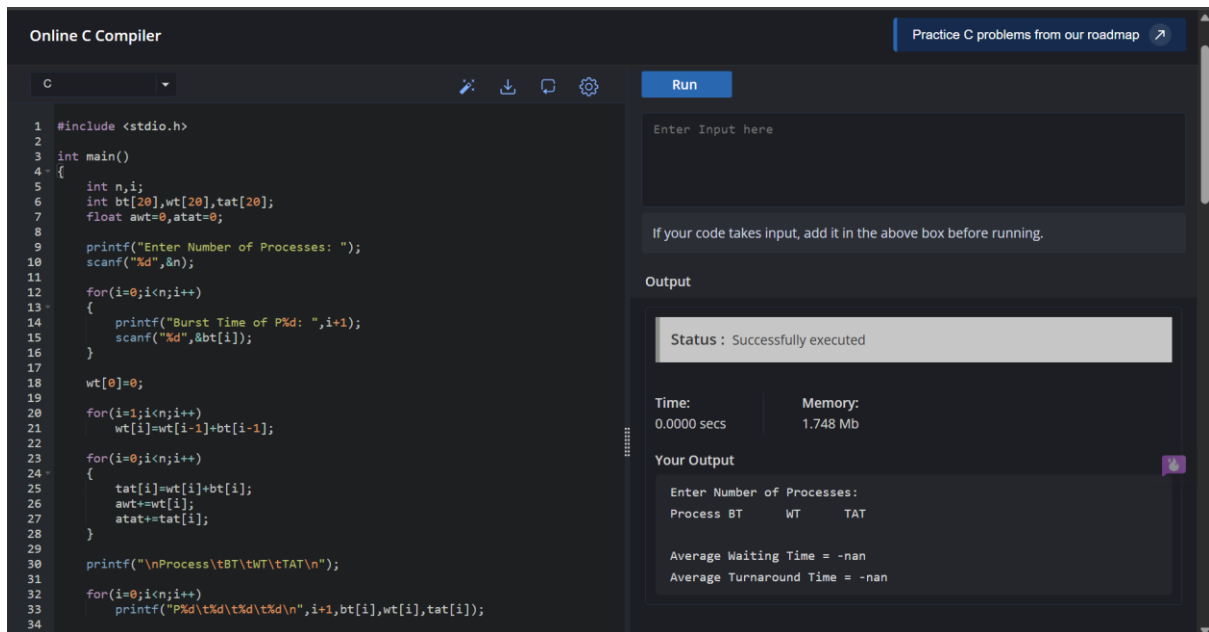


EXPERIMENT 03:

4. Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next.



The screenshot shows an online C compiler interface. On the left, a C program is written in a dark-themed editor. The program calculates burst times, waiting times, and turnaround times for a set of processes. On the right, the 'Run' button is visible. Below it, the 'Output' section shows the status 'Successfully executed', execution time (0.0000 secs), and memory usage (1.748 Mb). The 'Your Output' section displays the program's output, which includes a prompt for the number of processes and a table of process data.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n,i;
6     int bt[20],wt[20],tat[20];
7     float awt=0,atat=0;
8
9     printf("Enter Number of Processes: ");
10    scanf("%d",&n);
11
12    for(i=0;i<n;i++)
13    {
14        printf("Burst Time of P%d: ",i+1);
15        scanf("%d",&bt[i]);
16    }
17
18    wt[0]=0;
19
20    for(i=1;i<n;i++)
21        wt[i]=wt[i-1]+bt[i-1];
22
23    for(i=0;i<n;i++)
24    {
25        tat[i]=wt[i]+bt[i];
26        awt+=wt[i];
27        atat+=tat[i];
28    }
29
30    printf("\nProcess\tBT\tWT\tTAT\n");
31
32    for(i=0;i<n;i++)
33        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
34
35    printf("\nAverage Waiting Time = %.2f\n",awt/n);
36    printf("\nAverage Turnaround Time = %.2f\n",atat/n);
37 }
```

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.748 Mb

Your Output

Enter Number of Processes:

Process	BT	WT	TAT
P1	10	0	10
P2	5	10	15
P3	8	15	23
P4	3	23	26
P5	6	26	32

Average Waiting Time = 10.80
Average Turnaround Time = 21.40

PSEUDO CODE :

Start

Input number of processes

Input Burst Time

Sort processes by Burst Time

Waiting Time of first process = 0

For remaining processes

Waiting Time = Sum of previous burst times

Turnaround Time = Waiting Time + Burst Time

Calculate averages

Display results

Stop

Process	BT	WT	TAT
---------	----	----	-----

P3	2	0	2
----	---	---	---

P2	3	2	5
----	---	---	---

P1	5	5	10
----	---	---	----

Average Waiting Time = 2.33

Average Turnaround Time = 5.67

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,i,j,temp;
```

```
    int bt[20],wt[20],tat[20];
```

```
    float awt=0,atat=0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
        printf("Burst Time of P%d: ",i+1);
```

```

scanf("%d",&bt[i]);
}

for(i=0;i<n;i++)
{
    for(j=i+1;j<n;j++)
    {
        if(bt[i]>bt[j])
        {
            temp=bt[i];
            bt[i]=bt[j];
            bt[j]=temp;
        }
    }
}

wt[0]=0;

for(i=1;i<n;i++)
    wt[i]=wt[i-1]+bt[i-1];

for(i=0;i<n;i++)
{
    tat[i]=wt[i]+bt[i];
    awt+=wt[i];
    atat+=tat[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for(i=0;i<n;i++)
    printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);

printf("\nAverage Waiting Time = %.2f",awt/n);

```

```
printf("\nAverage Turnaround Time = %.2f",atat/n);
```

```
return 0;
```

```
}
```